

Needleless Transcutaneous Electrical Acustimulation: A Pilot Study Evaluating Improvement in Post-Operative Recovery

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BACKGROUND: Functional gastrointestinal disturbance occurs after abdominal surgeries and could last for an extended period of time in some cases. This study was designed (1) to evaluate the effects of needleless transcutaneous electrical acustimulation (TEA) on postoperative recovery, and (2) to investigate the mechanisms involving autonomic function in postoperative patients after removal of gastrointestinal cancers.

METHODS: Forty-two patients (33 male, age: 69.5 ± 1.5 years) scheduled for abdominal surgical removal of gastrointestinal cancers were randomized to TEA ($n=21$) and sham-TEA ($n=21$). TEA was performed via acupoints ST36 and PC6 1 h twice daily from the postoperative day (POD) 1 to day 3. Sham-TEA was performed at non-acupoints.

RESULTS: (1) TEA improved major postoperative symptoms by about 30%, including a reduction in time to defecation by 31.7% ($P < 0.01$ vs. sham-TEA), time to first flatus by 35.9% ($P < 0.001$), time to ambulation by 42.8% ($P < 0.01$), time to resuming diet by 26.5% ($P < 0.01$) and hospital stay by 30% ($P < 0.05$) as well as pain score by 50% ($P < 0.01$). (2) TEA significantly increased vagal activity ($P < 0.001$) and decreased sympathetic activity on POD 4 ($P < 0.001$) compared with POD 1 as well as the serum level of NE ($P < 0.05$). (3) The vagal activity, high frequency assessed from the spectral analysis of heart rate variability, was negatively correlated with time to resuming diet, whereas the sympathetic measurement, serum norepinephrine was positively correlated with time to resuming diet and time to flatus. (4) TEA but not sham-TEA decreased TNF- α by 17.4% from POD 1 to POD 4. (5) TEA was an independent predictor of a shorter hospital stay.

CONCLUSIONS: Needleless TEA improves major postoperative symptoms by enhancing vagal and suppressing sympathetic activities.

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INTRODUCTION

The postoperative disturbance is a frequent complication after abdominal surgery which is clinically characterized by delayed passage of gas and stool, resumption of oral feeding, postoperative ambulation and postoperative pain, and results in prolonged postoperative hospital stay and increased healthcare costs [1–3]. According to the Health Care Financing Administration (HCFA) data, Medicare patients who underwent surgical procedures

stayed in the hospital for a total of 1.82 million days (a mean of 11.3 days per patient), and cost healthcare industries approximately US \$1.75 billion during the year of 2000 [4]. It is estimated that the annual healthcare system costs will be reduced by 155 million dollars if the hospital stay for each patient is reduced by 1 day with the major abdominal colorectal surgeries in the United States [4].

Pharmacological treatments available for patients after major abdominal surgeries have showed limited effects [5]. Investigations

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as to whether prokinetics could be used to ameliorate postoperative ileus (POI) have come up with a negative conclusion. In 2008, a Cochrane review included 39 randomized controlled trials (RCTs) and investigated 15 systematic prokinetic drugs but could not identify any beneficial effects on time of first defecation and time of resuming diet after surgeries [6]. In the review, Alvimopan was shown to have a beneficial effect in several studies but the authors found there were many methodological biases. In a recent review, prucalopride was reported to shorten duration of POI and reduce the inflammatory response [7], but multi-center RCTs are needed to investigate which kind of patient would gain the most from this medicine [8].

It was well known that autonomic neurological dysfunction plays an important role in the progression of impaired gastrointestinal motility [9]. Early studies [10, 11] have shown that abdominal surgery increases sympathetic activity and decreases vagal activity. During the surgery, afferent signals are transmitted to the brainstem where they trigger an increased autonomic output to the neurons of the intermediolateral column of the thoracic cord, where sympathetic preganglionic neurons (releasing noradrenaline (NA)) are located [12–14]. In addition to this sympathetic overactivity, intestinal manipulation also activates resident macrophages in the intestinal muscularis externa which release proinflammatory cytokines and result in a long-lasting phase of POI [15].

Acupuncture is a traditional Chinese medicine that has been practiced for several thousands of years in China [16]. It has been used to treat functional disorders, including chemotherapy-induced nausea [17], postoperative nausea and vomiting [18, 19], irritable bowel syndrome, diarrhea [20] and constipation [21]. Electroacupuncture (EA) uses electrical pulses instead of manual manipulations to stimulate acupoints and appears to show more consistent and reproducible results [16]. In human studies, EA at ST36 has been shown to accelerate postoperative gastrointestinal function recovery [22]. In animal studies, EA is generally considered to enhance gastrointestinal motility by activating the vagus nerve and inhibiting sympathetic activity [23–26]. However, acupuncture and EA both use needles and this limits widespread applications since they have to be administered by healthcare professionals.

Transcutaneous electrical acustimulation (TEA) is a newly developed method of EA by replacing needles with surface electrodes. It has been shown to improve chemotherapy-induced gastric symptoms, cold meal-induced impaired gastric accommodation [27, 28] and chronic constipation [29]. TEA has been consistently shown to enhance gastrointestinal motility by improving vagal activity [9, 30]. However, it is unknown whether TEA might have a therapeutic effect on postoperative recovery. We hypothesized that major abdominal surgeries activate sympathetic nerve thus impair gastrointestinal (GI) motility and that TEA antagonizes surgery-induced sympathetic overactivity and therefore improves postoperative GI motility.

The aims of our study were to investigate the effects of TEA on postoperative recovery and to study the mechanisms involving autonomic function in patients with gastrointestinal cancers after major abdominal surgeries.

METHODS

Study participants

The study was performed from June 2016 through December 2016 at the Yinzhou People's Hospital. Forty-two patients with gastrointestinal cancers after open abdominal surgeries aged over 18 years were enrolled in this study. The sample size was estimated using previous data and power analysis. Forty subjects were required to detect a difference of at least 15% in time to first defecation after surgery between the TEA and sham-TEA groups with a conventional power of 80% and an α value of 0.05. The study protocol was approved by the ethical committee of Yinzhou People's Hospital. Inclusion criteria were as follows: (1) >18 years of age, (2) underwent an open abdominal surgery for cancers and (3) willing to sign a written informed consent. Exclusion criteria were: (1) diabetes, (2) multiple surgeries or single surgery with complications, (3) allergic to skin preparation and (4) familiar with acupoints and their functions.

Study design

Patients were recruited in the study immediately after surgery if all inclusion and exclusion criteria were fulfilled. Patients were randomized into two groups: TEA and sham-TEA. The postoperative management was the same between the two groups. Demographic data and surgical details were recorded. A blood sample was taken at 7:00 a.m. and placed on ice, and an electrocardiogram (ECG) was made from 8:30 a.m. to 9:00 a.m. on postoperative day (POD) 1 and POD 4. Each patient underwent two sessions of TEA or sham-TEA daily from POD 1 to POD 3. The visual analog scale (VAS) pain score, time of the first defecation, time of the first flatulence, time to resume diet and the length of the postoperative hospital stay were noted.

TEA and sham-TEA

Acupoints ST36 and PC6 were used for TEA [29–31]. ST36 is located at the place of four-finger-breadth measuring down from the outer eye of the knee between the fibula and the tibia, one-finger-breadth measurement beside the tibia. PC6 is located at the place of one-sixth of remote end and five-sixths of the proximal end of the connection stripe between the transverse wrist crease and cubital crease. A pair of electrodes were placed at bilateral ST36 and another pair of electrodes were placed at bilateral PC6 (Fig. 1). Each pair of electrodes were connected to a watch-size digital stimulator (SNM-FDC01, Ningbo Maida Medical Device Inc., Ningbo, China). The stimulation parameters for both stimulators were set at 2 s on and 3 s off, pulse width of 0.5 ms, pulse frequency of 25 Hz and amplitude of 2–10 mA (at the maximum level tolerated by the subject) based on previous studies [29–31]. The TEA treatment was given for 1 h, twice daily (9–10 a.m. and 4–5 p.m.), lasting for 3 days.

Sham-TEA was the same except that non-acupoints were used to replace ST36 and PC6. The sham-acupoint for PC6 was located at about 15–20 cm away from PC6 (up to the elbow and outside of the coastal margin of the forearm not on any meridian) and the sham-point for ST36 was located at 10–15 cm down from and to the lateral side of ST36 not on any meridian [29–31].

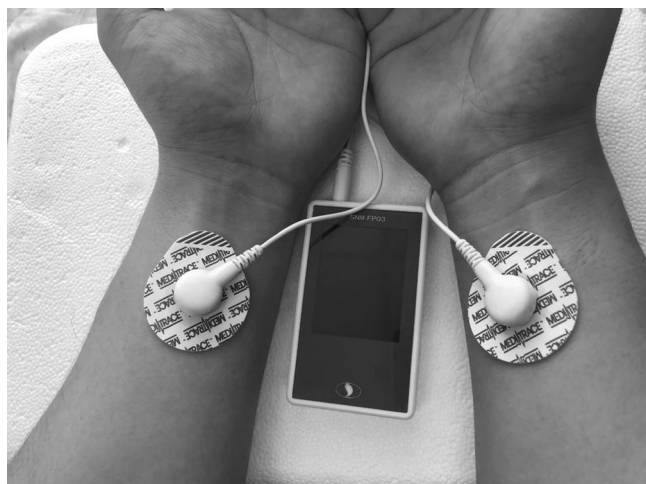


Fig. 1 Method of transcutaneous electrical acustimulation. One electrode was placed on acupoint PC6 and another at the same place but the other hand, as well as the ST36 acupoint. A watch-size stimulator was connected to the electrodes for stimulation

Outcome measurements

Main outcome measurements included time to the first defecation [22] and time to resume diet [32] after surgery. Other measurements included time to ambulation, length of postoperative hospital stay and the VAS pain score on the first 3 postoperative days. The VAS digital score was from 0 to 10; 0: painless; 1–3: mild and endurable pain; 4–6: moderate pain affecting sleep but still endurable; 7–10: severe pain, unbearable, affecting appetite and sleep. It was recorded by a caring physician who did not know the study design.

Assessment of autonomic function

The autonomic function was assessed by the spectral analysis of heart rate variability (HRV) derived from the ECG, described as follows: three locations in the chest/abdomen where ECG electrodes were to be placed (apical, the second intercostal of sternal right edge, and the intersection of right rib arch and clavicle midline) were marked, gently rubbed with sandy gel and wiped clean with a gauze [33]. Electrodes were placed over the treated skin, and the patient was asked to lie down and stay quiet. The ECG was recorded using a special amplifier with a recording frequency of 100 Hz (ECG-201, Ningbo Maida Medical Device Inc., Ningbo, China). The HRV signal was obtained by identifying the R peak and calculating the RR interval from the ECG using a custom-made software. The HRV signal was subjected to power spectrum analysis and the power in different bands was calculated. The power in the low-frequency (LF) band (0.04–0.15 Hz) reflects mainly sympathetic activity, whereas the power in the high-frequency (HF) band (0.15–0.50 Hz) represents purely parasympathetic or vagal activity. In this study, normalized values were utilized: that is, LF was calculated as $LF/(HF + LF)$ and HF was calculated as $HF/(HF + LF)$ [33, 34].

Blood draw and assay

A blood sample was taken at 7 a.m. on POD 1 and POD 4. About 5 ml of blood was drawn into a procoagulant tube placed on ice and centrifuged at 3000 rpm for 5 min at 8 a.m. (1 h later). The serum was divided into 3 portions, each about 0.5 ml and placed at -80°C for assay within 6 months. Human enzyme-linked immunosorbent assay (ELISA) kits (Roche Company, American, Cat. No. 0734-2014, Shanghai Shiyao Biological Technology Co., Ltd, China, Lot. No. 201704) were used for the analysis of interleukin-6 (IL-6) and noradrenaline (NE), respectively. Tumor necrosis factor- α (TNF- α) and interleukin-1 β (IL-1 β) were analyzed by human ELISA kits (Siemens Healthcare Diagnostics Products Limited Co., UK, Cat. No. LKNF1 and Cat. No. LKL11, respectively). All blood assays were performed blindly by a professional company (Shanghai Dian Medical Testing Laboratory).

Statistical analysis

The Shapiro–Wilk test was used to assess the distribution of major data, including time to defecation, flatus and diet, and the distributions were found normal and therefore parametric analyses were used. Data are expressed as mean $\pm$ standard error (SE). $P < 0.05$ was considered statistical significance. Student's *t*-test was performed to assess the difference in the measurement (time to defecation, time to flatulence, time to resume diet, length of postoperative hospital stay or VAS pain score) between sham-TEA and TEA group. Paired Student's *t*-test was used to compare the difference in serum NE, TNF- α , and IL-6, and HF (vagal activity) and LF/HF (sympathovagal ratio) on POD 1 and POD 4 in each group. Pearson's correlation analysis was performed to assess the correlation between the endpoint measures and other parameters. Multivariate analysis with multiple linear regression was used to identify independent predictors of outcome measures.

Time to defecation higher than 96 h was defined as the endpoint event. Comparison between endpoint and non-endpoint groups was performed using Chi-square Analysis, and P values of <0.05 were considered statistically significant. Clinic pathological parameters ($P < 0.15$) were tested by stepwise multivariate logistic regression analysis, and P values of <0.05 were considered statistically significant. The HF on POD 4 in patients with suspected delayed recovery was assessed and evaluated by receiver operating characteristic (ROC) curve. Data were analyzed by statistical software SPSS 19.0.

RESULTS

Characteristics of patients

A total of 42 patients were recruited and randomized into sham-TEA (21 patients) and TEA (21 patients) groups. Demographic data and surgical details are shown in Table 1. There were no significant differences in gender, age or body mass index (BMI) between the two groups (all $P > 0.05$). The two groups were similar in terms of surgical details and no significant difference was noted in the number of patients who underwent any particular type of surgery between the TEA and sham-TEA groups.

Table 1 Characteristics of patients and surgical details

Observation index	Sham-TEA	TEA
Patient (n)	21	21
Male (n (%))	15 (71.4)	18 (85.7)
Age (mean $\pm$ SE)	64.0 $\pm$ 2.6	68.0 $\pm$ 1.4
BMI (mean $\pm$ SE, kg/m ²)	20.0 $\pm$ 1.34	23.9 $\pm$ 1.99
Diagnosis, GC/CC/RC	16/2/3	15/1/5
Type of surgery, RNY/BI/BII/ CCR/ RCD/RCM	11/3/2/2/1	6/8/1/1/4/1
Surgical time, mean $\pm$ SE, min	197.4 $\pm$ 16.8	181.3 $\pm$ 10.0
Median blood loss(range), mL	100(0–200)	100(0–200)

BMI/ body mass index, GC/CC/RC gastric, colon, rectal cancer, BI Billroth I, BII Billroth II, RNY Roux-en-y, CCR colon cancer resection, RCD and RCM rectal cancer resection Dixon and Miles respectively

TEA improved postoperative recovery

TEA shortened the recovery time of clinic symptoms. Compared to sham-TEA, TEA shortened time to defecation by 31.7% ($P < 0.01$) and time to first flatus by 35.9% ($P < 0.001$) (Fig. 2a). Prolonged postoperative ileus as defined in a previous study [35] was noted in 5 patients in the Sham-TEA group but none in the TEA group (Chi-square analysis, $P < 0.05$). TEA reduced time to ambulation by 42.8% ($P < 0.01$), time to resuming diet by 26.5% ($P < 0.01$) and length of postoperative hospital stay by 30.2% ($P < 0.05$) in comparison of sham-TEA (Fig. 2b).

TEA reduced the VAS pain score on the postoperative Day 2 by 50.8% ($P < 0.01$) and Day 3 by 64.9% ($P < 0.001$) in comparison with sham-TEA (Fig. 2c).

Effects of surgeries and TEA on inflammatory cytokines

TEA but not sham-TEA decreased the serum level of TNF- α by 17.4% from POD 1 to POD 4 ($P < 0.05$) (Fig. 3a). The TNF- α on POD 4 was positively correlated with VAS pain scores on POD 2 ($r = 0.410$, $P < 0.01$) and POD 3 ($r = 0.402$, $P < 0.05$).

No significant difference was noted in the serum level of IL-6 between the TEA and sham-TEA groups (Fig. 3b); nor was there a difference between POD 1 and POD 4. The level of IL-6 on POD 1 was actually comparable to that in healthy subjects [36], suggesting that the surgical procedure did not alter IL-6.

The serum level of IL-1 β was too low to detect in any of the samples, suggesting that it was not elevated by surgery.

Mechanisms of TEA involving autonomic functions

TEA enhanced vagal activity. Vagal activity (HF) was increased from POD1 to POD 4 with TEA ($P < 0.001$) but not with sham-TEA ($P > 0.05$) (Fig. 4a). Similarly, the sympathovagal ratio (LF/HF) was decreased significantly with TEA ($P < 0.001$), but not with sham-TEA ($P > 0.05$) (Fig. 4b). TEA but not sham-TEA decreased the serum level of NE by 42.5% from POD 1 to POD 4 ($P < 0.05$) (Fig. 4c).

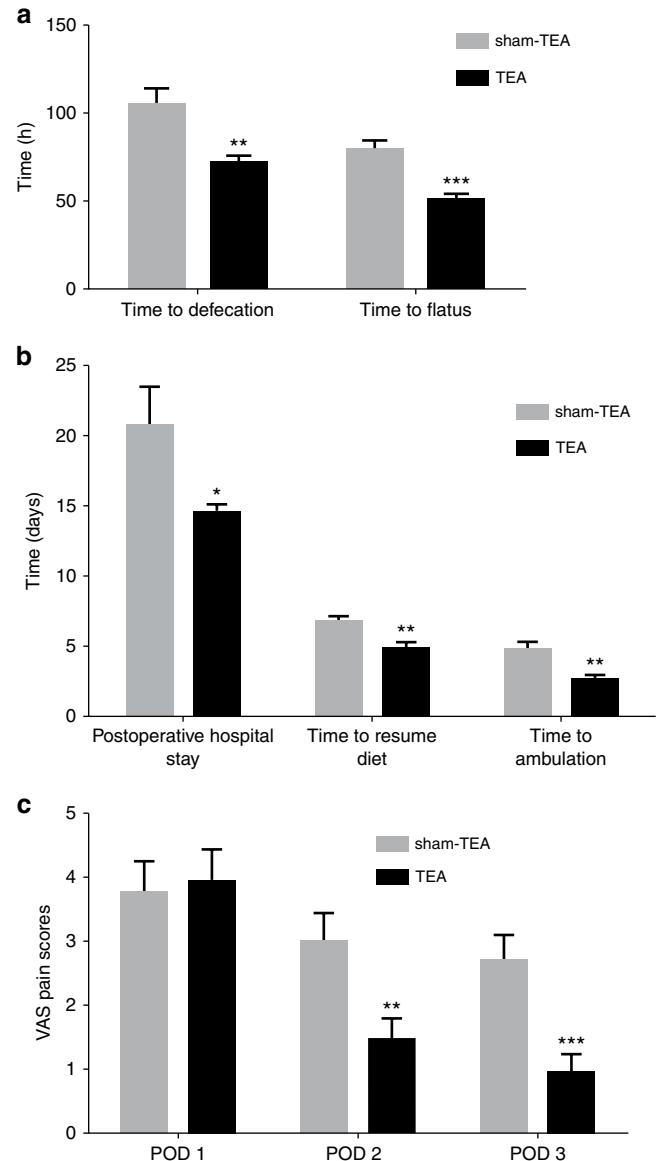


Fig. 2 Effects of TEA on postoperative clinical symptoms. TEA shortened time to defecation and time to flatus (a), as well as postoperative hospital stay, and time to resume diet and time to ambulation (b). TEA reduced postoperative pain on POD 2 and POD 3 (c). Vs. sham-TEA, * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; $n = 42$, analyzed by Student's t -test

The HF on POD 1 and 4 were negatively correlated with NE on POD 1 ($r = -0.353$, $P < 0.05$) and NE on POD 4 ($r = -0.328$, $P < 0.05$), respectively, as well as on both of two days ($r = -0.353$, $P < 0.01$), suggesting an agreement between these two measures of the autonomic function.

In addition to the direct effect of TEA on the vagal activity assessed by the HF, the HF was also found to be associated with a number of clinical outcome measures and even capable of predicting certain clinical outcomes as shown below:

Correlation of autonomic functions with clinic symptoms. The HF on POD 4 was negatively correlated with time to resume diet ($r = -0.407$, $P < 0.01$) (Fig. 5a), and time to resume diet was

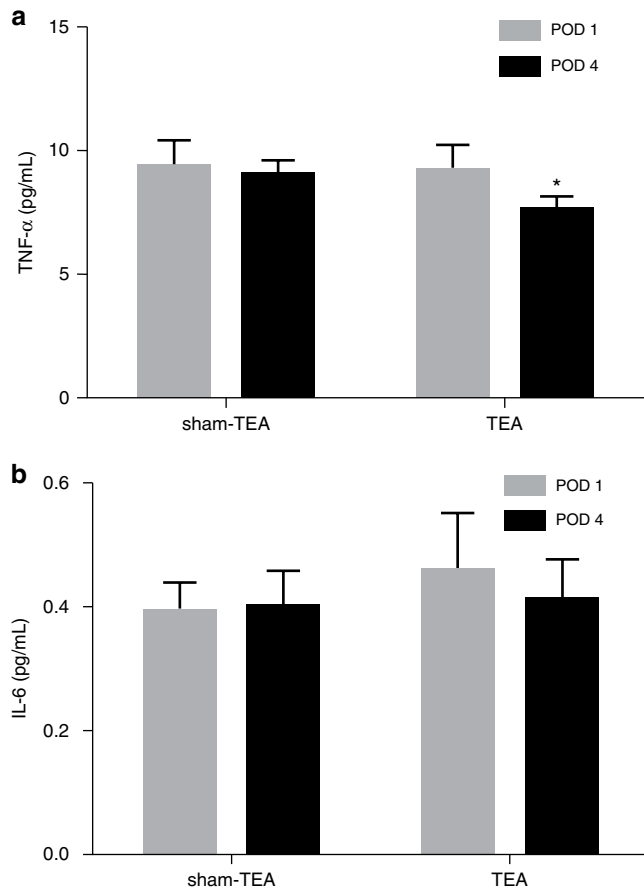


Fig. 3 Effects of TEA on inflammation. TEA decreased serum TNF- α . (a). TEA did not decrease serum IL-6 significantly (b). POD 4 vs. POD 1, * $P < 0.05$; $n = 42$, analyzed by paired t -test

positively correlated with time of first of defecation ($r = 0.431$, $P < 0.01$) (Fig. 5b). The NE on POD 4 was positively correlated with time to resume diet ($r = 0.479$, $P < 0.001$) and time to flatus ($r = 0.391$, $P < 0.01$). Also, NE on POD 1 was positively correlated with time to flatus.

Correlation of HF with IL-6. The HF on POD 1 and POD 4 was negatively correlated with IL-6 on POD 1 and 4 ($r = -0.221$, $P < 0.05$).

Correlation of NE with IL-6. The NE on POD 1 and 4 were positively correlated with IL-6 on POD 1 ($r = 0.570$, $P < 0.001$) and IL-6 on POD 4 ($r = 0.782$, $P < 0.001$), also on both 2 days ($r = 0.646$, $P < 0.001$).

HF as an independent predictor of outcome measures. Many confounding factors, such as age, sex, BMI, intraoperative blood loss, types of surgery, surgical time and TEA, might affect the duration of hospital stay. Their effects were analyzed by multiple linear regression. Interestingly, TEA was found to be the only independent predictor for a shorter duration of postoperative hospital stay (regression coefficient, -6.3 ; 95% confidence interval, -12.0 to -0.6 ; $P < 0.05$).

To determine whether the HF and some other parameters were capable of predicting clinical outcomes of the TEA therapy, multi-

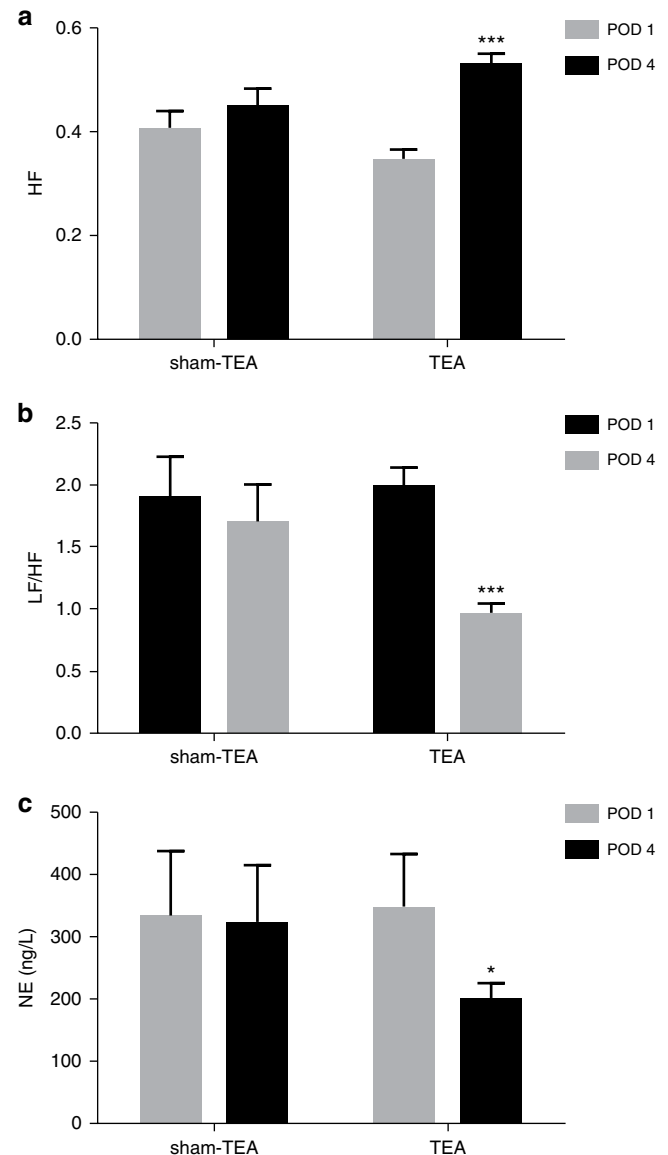


Fig. 4 Effects of TEA on autonomic function. TEA increased HF (a) and decreased LF/HF (b). TEA decreased serum NE (c). POD 4 vs. POD 1, * $P < 0.05$; *** $P < 0.001$, $n = 42$, analyzed by paired t -test

variate logistic regression analyses were performed in five variables that showed a significant or a trend of change with TEA: (1) BMI > 24 , (2) surgical approach (Roux-en-y or not-RNY), (3) time to ambulation > 4 , (4) HF4 (HF on POD 4) < 0.45 and (5) LF/HF4 (LF/HF on POD 4) > 1.5 (Table 2). It was found that HF4 < 0.45 was an independent risk factor predicting the occurrence of the endpoint event (time to defecation higher than 96h) in patients after open abdominal surgery (Table 3).

Further, the ROC curve was derived for the HF on POD 4 to predict a clinical outcome of "first defecation > 96 h" and the area under the curve was 0.695 (Fig. 6), suggesting that HF4 < 0.45 might be a predictor of long-time recovery in patients after open abdominal surgery.

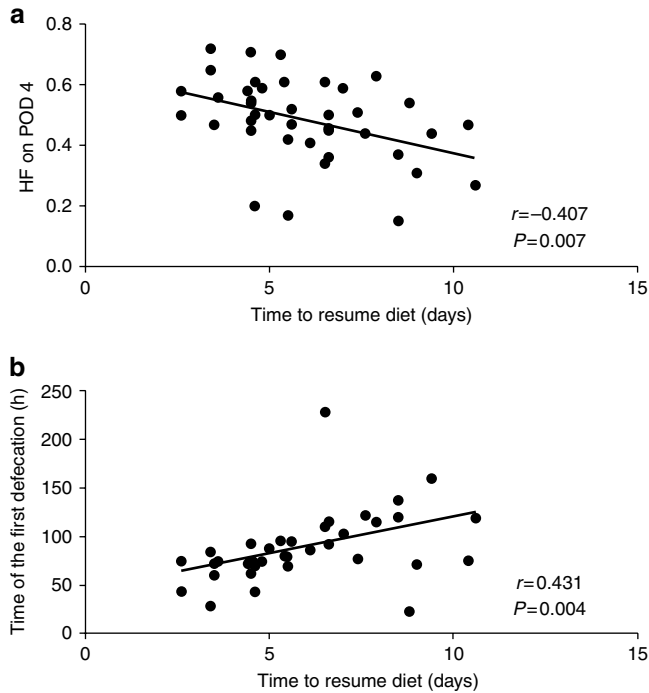


Fig. 5 Correlation between vagal activity and clinic symptoms. HF on POD 4 was negatively correlated with time to resume diet (a), and time to resume diet was positively correlated with time to defecation (b)

Table 2 Comparison of data between patients who reached and did not reach the observation endpoint

Observation index	Non-endpoint group (n=33)	Endpoint group (n=11)	P value
Age >65 (%)	(17) 54.8	(7) 63.6	0.612
Male (%)	(25) 80.6	(7) 63.6	0.256
BMI >24 (%)	(10) 32.3	(1) 9.1	0.133
Disease (GC, not GC) (%)	(22) 71	(9) 81.8	0.482
Surgical approach (RNY, not RNY) (%)	(10) 32.3	(7) 63.6	0.069
Time of surgery >3 h (%)	(14) 45.2	(4) 36.4	0.612
Time to resume diet >7 day (%)	(22) 71	(8) 72.7	0.912
Time to ambulation >4 day (%)	(11) 35.5	(7) 63.6	0.105
VAS1 >3 (%)	(15) 48.4	(7) 63.6	0.384
VAS2 >3 (%)	(5) 16.1	(3) 27.3	0.419
VAS3 >3 (%)	(3) 9.7	(3) 27.3	0.152
HF1 <0.45 (%)	(26) 83.9	(7) 63.6	0.160
HF4 <0.45 (%)	(5) 16.1	(7) 63.6	0.003
LF/HF1 >1.5 (%)	(20) 64.5	(5) 45.5	0.268
LF/HF4 >1.5 (%)	(4) 12.9	(5) 45.5	0.024

BMI body mass index, VAS visual analog scale, LF1 low frequency on postoperative day 1, HF4 high frequency on postoperative day 4

Table 3 Multivariate logistic regression analysis on defecation progression to the observation endpoint

Observation index	OR value	P value
BMI >24	1.087	0.297
Surgical approach (RNY, not-RNY)	5.515	0.060
Time to ambulation >4 day	0.673	0.412
HF4 <0.45	12.377	0.006
LF/HF4 >1.5	0.248	0.618
OR odds ratio		

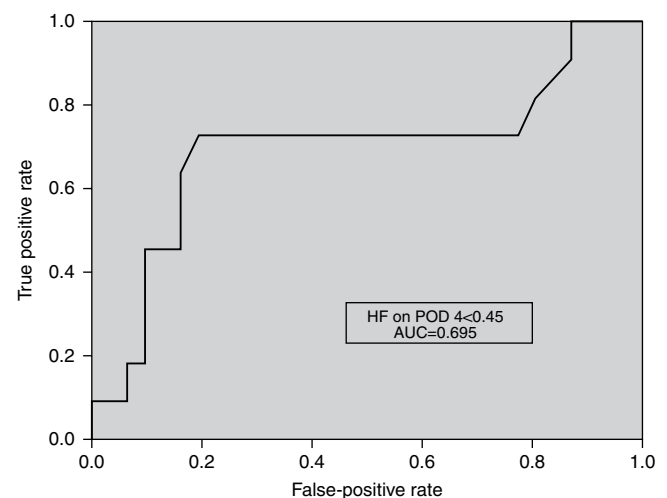


Fig. 6 ROC curve of HF test on postoperative day 4

DISCUSSION

In this study, we found that: (1) TEA improved major postoperative symptoms, including a reduction in time to first defecation, time to first flatus, time to ambulation, time to resuming diet and hospital stay as well as pain score. (2) TEA but not sham-TEA decreased the serum level of TNF- α . (3) TEA significantly increased vagal activity and decreased sympathetic activity as well as the serum level of NE. (4) The vagal activity, HF, assessed from the spectral analysis of HRV was negatively correlated with time to resuming diet, whereas the sympathetic measurement, serum norepinephrine was positively correlated with time to resuming diet and time to flatus. (5) TEA was an independent predictor for a shorter hospital stay.

While the ameliorating effect of needle-based EA on postoperative recovery had been reported [22] it was unclear whether needleless TEA would have a similar therapeutic effect on postoperative recovery. We designed a prospective randomized sham-controlled study to investigate the effects of TEA for postoperative recovery and its mechanisms involving autonomic functions. The newly developed method of TEA, which is needleless and can be self-administrated, is expected to be well accepted by postopera-

tive patients who are under close monitoring of various functions and medications. In this study, there was no significant difference in gender, age, BMI, severity of disease or surgical procedures between the two groups. In order to avoid placebo effects, the sham-controlled group was treated with the same electrical stimulation at the non-acupoints [9, 28, 30].

Substantial reductions were noted in major postoperative symptoms with the noninvasive treatment of TEA. TEA shortened time to defecation and time to flatulence by more than 30%, and time to resume diet by 26.5% in comparison of sham-TEA. In addition, TEA reduced length of postoperative hospital stay by 30.2% and was found to be an independent predictor of a shorter hospital stay. In a previously clinic study, needle-based EA was reported to reduce time to defecation by 29.6% and the hospital stay by 23.5% compared with control in patients after laparoscopic surgery for colorectal cancer [22]. They also found that EA was an independent predictor of shorter duration of postoperative recovery. In comparison with the previous EA therapy, the TEA method used in this study has an obvious advantage of being needleless: the therapy can be administered by anyone including the patient. To the best of our knowledge, this was the first study to demonstrate the therapeutic effects of needleless TEA for postoperative recovery.

The best indication of GI motility recovery is the combined measure tolerance of solid food and passage of stool [32]. Accordingly, we speculated that these improvements were probably attributed to improvement in GI motility. GI motility was not assessed in this study due to invasiveness of the procedure and severity of the disease and surgical procedures in these patients. However, previous EA and TEA studies have suggested a possible role of improved gastric, intestinal and colonic motility. EA at ST36 significantly accelerated gastric emptying of solids in normal rats [37] and in a rodent model of delayed gastric emptying induced by restraint stress [38]. Ouyang et al. [39] reported that EA at PC6 and ST36 significantly improved gastric emptying of liquid in dogs with delayed gastric emptying induced by duodenal distention. EA at PC6 and ST36 improved gastric emptying of solids measured by scintigraphy in patients with gastroparesis [40]. Improvement of intestinal motility with EA has been reported in animals [38, 41, 42]. Enhancement of colonic motility with EA or TEA has been investigated in both animals and humans. In conscious rats, contractility of the distal colon and colonic transit were significantly increased by EA [43]. Recent studies reported that EA at ST36 improved colonic contractions and transit impaired by rectal distention [44] and surgery [23] by vagal activity enhancement. Similar findings were also investigated in TEA studies. Zhang et al. [30] found that the TEA treatment at ST36 could improve the stress-induced gastric slow wave impaired by balancing sympathovagal activities. Huang et al. [28] found that TEA at ST36 and PC6 improved gastric accommodation, gastric slow wave damage and dyspeptic symptoms induced by cool meal in healthy volunteers. TEA has been shown to improve chronic constipation [29]. These previous consistent and positive findings support a prokinetic mechanism of TEA involved in the present clinical study.

Another noticeable finding of this study was that TEA reduced the VAS pain score by more than 50% on POD 2 and POD 3 in comparison of sham-TEA. Previous studies have suggested that EA blocks pain by activating a variety of bioactive chemicals through peripheral, spinal and supraspinal mechanisms. These include opioids which reduce proinflammatory cytokines and norepinephrine [45]. The analgesic effect of TEA observed in the current study was also believed to be attributed to these previously reported opioid and sympathetic mechanisms.

The effects of the surgery on postoperative gastrointestinal disturbance observed in this study might be secondary to changes in autonomic functions. It was well known that the autonomic neurological dysfunction plays an important role in the progression of impaired gastrointestinal motility [9]. Early studies [10, 11] have shown that surgery increases sympathetic activity and decreases vagal activity. In the present study, using the spectral analysis of the HRV, a noninvasive and simple method which has been well documented to be an accurate and reliable method for the quantitative evaluation of parasympathetic activity and sympathovagal balance [46, 47], we found that TEA increased vagal activity (HF) and restored the sympathovagal balance (LF/HF). Similar autonomic mechanisms were also reported in other EA and TEA studies. EA at ST36 and PC6 with appropriate parameters has been consistently shown to activate the vagus nerve and decrease sympathovagal balance [23–26, 44]. The needleless TEA has also been reported to be able to balance the sympathovagal activities under various conditions of sympathetic overactivity [9, 30, 48]. In the present study, in addition to the assessment of autonomic functions via the spectral analysis of HRV that provided accurate measures of parasympathetic/vagal activity and the sympathovagal ratio, we also measured the serum level of NE that is an indicator of sympathetic neural activity [49, 50]. Sympathetic neural reflex has been proposed as one of the predominant mechanism of POI [51, 52]. Fukuda et al. [53] reported that sympathetic-adrenergic pathway mediated entire GI tract motility at the early and late stages of postoperative disturbance and this pathway could be inhibited by α -2-adrenoceptor agonist. While the exact neural pathway was not investigated in the current study, previous animal studies [23, 54] have reported that: (1) EA at ST36 activates nucleus tractus solitaries by activating somatic afferent nerves, demonstrating the involvement of the central nervous system; (2) the afferent signal stimulates the dorsal motor nucleus of vagus, resulting in increased vagal efferent flow to visceral organs; and (3) the afferent signal is conveyed to other parts of the central nervous system, resulting in a reduced sympathetic outflow. Intuitively, the effects of TEA on postoperative recovery might follow the same pathways as EA: TEA activated somatic nerves that conveyed a signal to the brain via the spinal afferent and the brain processed the signal and produced an enhanced vagal efferent to the GI tract, resulting in improvement in motility and reduction in postoperative symptoms.

The inflammatory response has been proposed as another predominant mechanism of POI [51, 52]. In the current study, TEA decreased serum TNF- α on POD 4 in comparison of POD 1. In a previous animal study, needle-based EA decreased the plasma concentration of TNF- α in a rodent model of 2,4,6-trinitrobenzene

sulfonic acid (TNBS)-induced colitis via activating vagus nerve [55]. In a clinical study, EA decreased the serum level of TNF- α in patients with rheumatoid arthritis [56]. De Jonge et al. [51] reported that intestinal manipulation during major surgery increased inflammatory infiltrates and then triggered the inhibitory enterogastric neurons pathway which resulted in POI in mice. Later, De Jonge et al. [57] found that vagal activity attenuated the surgery-induced inflammation and POI and decreased the secretion of TNF- α via $\alpha 7$ subunit-mediated Jak2-STAT3 (Janus kinase/signal transducer and activator of transcription 3) activation in intestinal macrophages in a mouse model of intestinal manipulation [57]. Recently, Matteoli et al. [58] suggested that vagal activity reduced intestinal inflammation and ameliorated POI through $\alpha 7$ nicotinic acetylcholine receptor activity in the intestine muscularis-resident macrophages in a mouse model of intestinal manipulation. Accordingly, we speculate that TEA decreased TNF- α by activating cholinergic anti-inflammatory pathway due to its enhancement in vagal activity. Furthermore, a number of recent studies have shown the role of inflammation in animal models of POI and suggested that improvement in inflammation may lead to restoration of gastrointestinal motility. Farro et al. [59] reported a critical role of monocyte-derived macrophages in restoring intestinal hemostasis after surgical trauma. Deng et al. [60] reported an ameliorating effect of acupuncture on POI by suppressing inflammation and protecting interstitial cells of Cajal, resulting in improved intestinal transit.

Surprisingly, the current study failed to show an increase of serum IL-6 or IL-1 β . The level of IL-6 measured in this study was comparable to that in healthy controls [36], suggesting that IL-6 was not sufficiently elevated due to the surgeries. Similarly, the level of IL-1 β was found undetectable as reported in healthy controls [61].

Other interesting results of this study were the correlations of the postoperative symptoms and the autonomic functions. The vagal activity (HF) was negatively correlated with time to resume diet and serum NE was positively correlated with time to resume diet and time to flatus. These were interesting correlation findings that had never been reported previously suggested again the possible mechanisms involved in the improvement of postoperative symptoms with TEA: TEA improved GI motility by suppressing sympathetic activity and enhancing vagal activity, resulting in improvement in postoperative symptoms. Also, we found a negative correlation between the vagal activity assessed from the spectral analysis of HRV and the serum level of NE, demonstrating that these two measurements were consistent and complementary with each other [62, 63].

Furthermore, we discovered that HF <0.45 was an independent risk factor for predicting the occurrence of longer postoperative recovery or postoperative ileus. Our findings indicated that if the vagal activity, HF assessed from the spectral analysis of HRV, was lower than 0.45 on POD 4, the patient would need more time to recover from the surgery. This suggests that in postoperative patients, the HF should be monitored constantly and more aggressive management (prokinetic therapy) should be considered if this value is low.

There were several limitations in our study. First, patients had a long duration of hospital stay in comparison with that in western countries. This was attributed to the social background, such as culture and tradition, and hospitalization costs. In China, due to relatively low costs of hospital stay and serious concerns of both patients and surgeons on major invasive surgical procedures and postsurgical recovery, patients stay at hospital much longer to ensure a complete recovery. Second, symptoms of upper abdominal fullness, nausea and vomiting after the surgery were not recorded. However, due to the fact that very few patients had these symptoms, it was unlikely to observe a significant difference in these symptoms between the sham-TEA and TEA groups. Third, this was a single center study and some bias might be present in the study.

In conclusion, needleless TEA at PC6 and ST36 is able to accelerate postoperative recovery after open abdominal surgeries in patients with GI cancers by enhancing vagal activity. Needleless and self-administrable TEA may be considered as an easy-to-implement and low-cost complementary therapy to improve postoperative recovery.

CONFLICT OF INTEREST

Guarantor of the article: The first author Bo Zhang is accepting full responsibility for the conduct of the study.

Specific author contributions: Conceiving and designing the experiments: Liang Zhu, Jiande DZ Chen, Bo Zhang. Performing acustimulation treatment and heart rate variability test: Bo Zhang. Recording symptoms and sampling blood: Feng Xu, Mingyuan Zhang, Kehui Tong. Analyzing data: Bo Zhang, Pingping Hu, Gang Ma, Yuemei Xu. Contributing reagents/materials/analysis tools: Feng Xu, Pingping Hu, Liang Zhu, Jiande D.Z. Chen. Writing and revising paper: Liang Zhu, Jiande DZ Chen, Bo Zhang.

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Potential competing interests: None.

Study Highlights

WHAT IS CURRENT KNOWLEDGE

- ✓ Gastrointestinal disturbance is a frequent complication after abdominal surgeries.
- ✓ Electroacupuncture has been shown to accelerate postoperative gastrointestinal function recovery.
- ✓ Transcutaneous electrical acustimulation has been shown to enhance gastrointestinal motility.

WHAT IS NEW HERE

- ✓ TEA significantly increased vagal activity and decreased sympathetic activity as well as the serum level of NE.
- ✓ The vagal activity was negatively correlated with time to resuming diet.
- ✓ TEA decreased the serum level of TNF- α .
- ✓ TEA was an independent predictor for a shorter hospital stay.

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